# Ethan Mahajan

Mechanical Engineering Student

emahajan7@gatech.edu | Dacula, GA | ethan-mahajan | EthanMahajan

## Professional Summary

- Mechanical engineering student with experience in CAD design, additive manufacturing, electromechanical prototyping, and freelance engineering consulting. Completed 20+ client projects while independently designing and building a custom CoreXY 3D printer.

## Education

**Georgia Institute of Technology**, Mechanical Engineering — Atlanta, GA
Aug 2026 – May 2029

- GPA: 4.00
- Relevant Coursework: Principles & Applications of Engineering Materials; Engineering Graphics

**Mill Creek High School**, STEM — Hoschton, GA
Aug 2022 – May 2026

- Weighted GPA: 4.63; SAT: 1540 (Top 1% Nationally)
- AP Scholar with Distinction; National Honor Society
- Physics and Math Scholar Award (1 of 700 Students)
- Relevant Coursework: AP Physics; AP Calculus; AP Chemistry

## Skills

**CAD & Design:** Fusion 360, Solidworks, Onshape, FreeCAD, Technical Drawings, Design for Manufacturing

**Manufacturing & Fabrication:** FDM 3D Printing, Rapid Prototyping, Mechanical Assembly, Woodworking, Metalworking, Basic Metrology

**Electronics & Hardware:** KiCad Schematic Design (PCB Layout Pending), Soldering, Wiring, Crimping, Multimeter Diagnostics, CAN Bus, Connector Assembly

**Machinery & Tools:** FDM 3D Printers, Laser Engravers, Milling Exposure, & Wood Saws

**Software & Systems:** Microsoft Office, Google Workspace, Github, Windows, Linux, MacOS, Klipper, Java, Python, & Markdown

**Communication & Project Work:** Technical & Client Communication, Team Collaboration, Project Management & Documentation, Online Tutoring, Lesson Planning, Time Management

## Projects

**FDM 3D Printer Design, Prototyping, & Revising** — June 2024 – July 2026

Created a custom CoreXY 3D printer by overhauling an older machine.

- Designed a frame and motion system in CAD to meet mechanical and thermal specifications.
- Verified component tolerances and clearances before fabrication.
- Fabricated a prototype using metalworking machines and additive manufacturing.
- Configured Kalico Klipper firmware and performed exploratory motion tests; the printer is operational.
- Installed CAN-bus toolhead electronics to compartmentalize for easier maintenance.
- Revised frame support and used vibration analysis and input shaping to investigate ringing; repeatable print quality remains under refinement.

**Air Purifier** July 2026 – present

Controller design for a proposed smart air purifier using a 120 mm fan and particulate-matter sensor.

- Designed a control concept for automatic fan-speed adjustment and filter replacement detection.
- Completed electrical schematics in KiCad; PCB layout and physical validation remain pending.

**Laser-Engraved Chess Board** Aug 2026 – present

In-progress Invention Studio project combining CAD, laser engraving, and planned CNC-machined metal pieces.

- Developed a quickly manufacturable board design and planned low-center-of-mass pieces for stability.
- Fabricated the laser-engraved playing surface and personalized reverse; CNC-machined metal pieces remain planned.

## Experience

**Fiverr**, Freelance CAD Designer & Manufacturing Consultant — Remote

Completed 20+ custom engineering contracts involving CAD design, reverse engineering, and small-scale manufacturing. — Feb 2024 – Oct 2025, 1 year 9 months

- Maintained a 4.6-star client rating across completed engagements.
- Designed and iterated an accessibility device, including a tie-tying assistance tool.
- Reverse-engineered a broken replacement component from reference materials and technical drawings.
- Manufactured a low-volume production run of 30 tailored tablet holders.

**Shape Machine**, Social Media & Web Design Contributor — Atlanta, GA

Work with an early-stage startup developing shape-computation tools and supporting public-facing digital communication. — Sept 2026 – present, 2 months

- Launched Shape Machine's digital presence through social-media outreach.
- Collaborate with an interdisciplinary team on the existing Shape Computation Lab website through content edits, corrections, and template fixes.
- Contribute GitHub documentation and website fixes through the team's enterprise repository.

**Wyzant**, Freelance Private Tutor — Remote

Work directly with students of different ages to schedule lessons, prepare instructional materials, and tailor tutoring to each student's needs. — May 2026 – Aug 2026, 4 months

- Tutored approximately 100 hours while maintaining a 5-star rating across approximately 70 reviews.
- Schedule lessons directly with students and prepare materials that clarify each lesson's subject matter.
- Developed online tutoring, time-management, lesson-planning, and student-communication skills.

**Mathnasium of Dacula**, Mathematics Instructor — Dacula, GA

Instructed students from foundational arithmetic through AP Calculus while adapting explanations to individual learning speeds. — Aug 2024 – Aug 2025, 1 year 1 month

- Instructed up to five students simultaneously in a fast-paced learning environment.
- Translated complex mathematical concepts into clear, step-by-step explanations.
- Supported student confidence and academic progress across diverse grade levels.

**Mercor & Handshake AI**, AI Data Trainer & Labeler — Remote

Analyze AI-generated artifacts and author prompts to provide structured feedback on model quality. — May 2026 – present, 7 months

## Involvement

**Technology Student Association**, Treasurer (TSA) — Hoschton, GA
Aug 2024 – May 2026

Managed club finances, supported engineering competition activities, and helped coordinate member communication.

- Managed club finances and structural budgeting alongside faculty advisors.
- Organized community fundraising initiatives, including a local movie-night event.
- Organized and delivered presentations about upcoming events and member opportunities.
- Competed at the state level as part of a biotechnology team.

**Osborne Middle School**, Volunteer Tutor — Aug 2025 – May 2026

Provided volunteer tutoring support to middle-school students.

## Languages

**English**
Native

**German**
Conversational

## Interests

**Advanced Manufacturing & Research:** Additive Manufacturing, Rapid Prototyping, Robotics & Electromechanical Systems, Hardware & Electronics, Experimental Design, Applied Engineering Research

**Systems, Software & AI:** Programming, Operating Systems (Windows, macOS, Linux), Engineering Software, Local & Agentic AI Experimentation

**Design, Craft & Personalization:** Personalized Product Design, Everyday Problem Solving, Woodworking, Culinary Experimentation

**Outdoors, Strategy & Learning:** Hiking & Outdoor Exploration, Chess, Continuous Technical Learning